



5.4

DATABASE BACKUP –

TYPES OF FAILURES,

CAUSES OF FAILURES,

DATABASE BACKUP INTRODUCTION,

TYPES OF DATABASE BACKUPS- PHYSICAL AND LOGICAL



TYPES OF FAILURES

- **Hardware failure**
 - **System crashes**
 - **Media failure**
- **Software failure**
 - **System crashes**
 - **Application software errors**
 - **Statement failure**



TYPES OF FAILURES

- **System crashes**
 - **A system crash usually refers to any kind of bugs or hardware malfunction in the operating system or the database software.**
 - **It can bring the processing of transaction to a halt and can even cause the loss of content residing on volatile storage such as main memory, cache memory, RAM, etc.**



TYPES OF FAILURES

- **Media failure**
 - **Media failures are caused by a head crash or unreadable media.**
 - **These types of data failures are considered one of the most serious because it is possible for entire data loss.**
 - **Media failures usually leave database systems unavailable for several hours until recovery is complete.**



TYPES OF FAILURES

- **Application software errors**
 - **When the resource limit is exceeded, bad input, logical or internal errors occur, or any other factors related to the application software is compromised, transactions can fail giving way to database failure.**



TYPES OF FAILURES

- **Statement failure**
 - **As we know, the database reflects the output based on the SQL queries or statements. So, a statement failure, as its name suggests, can be referred to as the inability of the database system to execute the given SQL statement**



CAUSES OF FAILURES

- **System crashes**
- **User Errors.**
- **Carelessness**
- **Sabotage (internal corruption of data)**
- **Media failure**
- **Application software errors**
- **Statement failure**
- **Network failure**
- **Natural Physical Disasters.**



INTRODUCTION TO BACKUP

- ❑ In case, any kind of failure strikes databases, a proper plan and a way of recovery from that failure is needed, beforehand.
- ❑ A good backup can indeed be the best way to recover from most of the failures, especially when the data is too critical or important.
- ❑ Backup is a **procedure** for making extra copies of data for the purpose of restoration in case of loss or damage.
- ❑ A backup is a safeguard against unexpected data loss and application errors.
- ❑ Backups are categorized into:
 - ✓ **Physical backups**
 - ✓ **Logical backups**



PHYSICAL BACKUPS

- **Copies of physical database files and storing database information to some other location, whether on disk or some offline storage such as tape.**
- **Two ways to perform physical backup:**
 - **Recovery Manager (RMAN) utility.**
 - **Operating system utilities.(user-managed backup)**
- **Physical backups are the foundation of any sound backup and recovery strategy.**



LOGICAL BACKUPS

- **Contain logical data (for example, tables and stored procedures)**
- **Extract with an Oracle utility (Export utility) and stored in a binary file.**
- **Logical backups used to supplement physical backups.**
- **Logical backups are a useful supplement to physical backups in many circumstances but are not sufficient protection against data loss without physical backups.**



BACKUP TERMINOLOGY

- **Hot Backup / On-line**
- **Cold Backup / Off-line**
- **Off-Site**
- **Data Compression**
- **Data Deduplication**
- **Data Encryption**
- **Full Backup**
- **Incremental Backup**
- **Differential Backup**



TYPES OF BACKUP

- A **full backup** (level 0) is a complete copy of a partition.
- An **incremental backup** (level 1) is an archive of only the files that have changed since the last full backup.
- A **differential backup** (level 2, 3, etc) is an archive of only the file that have changed since the last backup (not necessarily full backup).

Backup	Sun (F)	Mon	Tue	Wed	Thu	Fri	Sat
Full	2TB	2TB	2TB	2TB	2TB	2TB	2TB
Incr.	2TB	1GB	1.2GB	1.6GB	1.9GB	2.3GB	2.8GB
Diff	2TB	1GB	0.2GB	0.4GB	0.3GB	0.4GB	0.5GB



CREATING OFFLINE (COLD) BACKUPS

1. Shut down database instance
2. Copy specific database files to alternate disk location
 1. Parameter file
 2. Control files
 3. Data files
 4. Not redo logs
3. Restart database instance



CREATING ONLINE (HOT) BACKUPS

1. Place database instance in ARCHIVELOG mode
2. Back up control file
3. Back up data files
4. Instruct database to write current contents of redo log files to archive files
 - Create backup copies of archive log files



ORACLE BACKUP STRATEGIES

- Cold backups
 - Simplest to perform
 - Satisfactory for databases whose:
 - Contents do not change very much
 - Users can tolerate downtime while DBA creates cold backup files
- Hot backups
 - Complex to perform
 - Better suited for dynamic, mission-critical databases



5.5

DATABASE RECOVERY – RECOVERY CONCEPT, RECOVERY TECHNIQUES- ROLL FORWARD, ROLLBACK



DATABASE RECOVERY

- Process of restoring database from either cold or hot backup files
- Necessary after database failure
- Cold backup recovery:
 - Shut down database
 - Restore backed up control file and datafiles to database server
 - Restart database



DATABASE RECOVERY (CONTINUED)

- Hot backup recovery:
 - Recover database to point at which last hot backup created
 - Use control trace file to generate new copy of control file
 - Restore backup copies of datafiles
 - Perform media recovery



ROLLBACK & ROLLFORWARD

- There are two modes to database recovery, ROLLBACK and ROLLFORWARD.
- ROLLBACK is the most common form of recovery.
- The other form of database recovery is ROLLFORWARD.
- If a hard disk fails, then a ROLLFORWARD is performed to restore the database to the point in time of the disk failure. The logs are kept because they would need to be applied to a full backup of the database to restore the database to the point of disk failure.



ROLLBACK

- **Rollback transaction is a transaction which rolls back the transaction to the beginning of the transaction.**
- **The transaction can be rolled back completely by specifying the transaction name in the Rollback statement or to cancel any changes to a database during current transaction.**
- **It is permissible to use before Commit transaction.**



ROLLFORWARD

- **It is the process of recovering a database by applying different transactions that recorded in the database log files.**
- **It is nothing but re-doing the changes made by a transaction i.e after the committed transaction and to over write the changed value again to ensure consistency.**



QUESTIONS

